

ServiceNow Metro Ticket Generating System

Project Documentation

Platform: ServiceNow

Project Type: Service Catalog based Metro Ticket Booking System

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1. Project Overview

The ServiceNow Metro Ticket Generating System is designed to simplify metro ticket booking through the Service Catalog. Users can select stations, choose a journey type and passenger count, calculate the applicable fare, select a payment mode, and generate a QR-based metro ticket.

2. Objectives

- Provide a simple metro ticket booking interface.
- Allow users to select source and destination stations.
- Support single and return journeys.
- Allow selection of number of passengers.
- Calculate fare dynamically based on route and passenger count.
- Provide different payment mode options.
- Generate a QR code for the metro ticket.
- Preserve the ServiceNow configuration through an Update Set.

3. Technologies and ServiceNow Components

- ServiceNow Service Catalog
- Service Portal
- Catalog Client Scripts
- Catalog UI Policies
- Service Portal Widget
- JavaScript
- Update Sets
- GitHub

4. Catalog Item

Catalog Item: Book A Metro Ticket

Variables: Starting From, Going To, Type of Journey, No of Passengers, Amount for Single Journey, Amount Including Return, Mode of Payment, Enter Payment Mode.

5. Supported Stations

Ameerpet, Madhapur, LB Nagar, Uppal Stadium, Jubilee Hills, Panjagutta, Kukatpally.

6. Journey and Passenger Options

Journey Types: Single Journey and Return Journey.

Passengers: 1 to 5.

Payment Modes: UPI, Card, Others.

7. Dynamic Fare Calculation

For a single journey: Fare = Route Fare × Number of Passengers.

For a return journey: Fare = Route Fare × 2 × Number of Passengers.

The fare logic supports reverse routes as well.

Route	Fare
Ameerpet - Panjagutta	■20
Ameerpet - Madhapur	■30
Ameerpet - Jubilee Hills	■20
Ameerpet - Kukatpally	■30
Ameerpet - LB Nagar	■50
Ameerpet - Uppal Stadium	■60
Panjagutta - Jubilee Hills	■20
Panjagutta - Madhapur	■30
Panjagutta - Kukatpally	■40
Panjagutta - LB Nagar	■50
Panjagutta - Uppal Stadium	■60
Jubilee Hills - Madhapur	■30
Jubilee Hills - Kukatpally	■40
Jubilee Hills - LB Nagar	■50
Jubilee Hills - Uppal Stadium	■60
Madhapur - Kukatpally	■40
Madhapur - LB Nagar	■50
Madhapur - Uppal Stadium	■50
Kukatpally - LB Nagar	■60
Kukatpally - Uppal Stadium	■70
LB Nagar - Uppal Stadium	■30

8. Catalog Client Scripts

Fare auto-calculation: Calculates the fare when the source, destination, journey type, or passenger count changes.

QR Generation: Opens the Metro QR Widget during ticket submission and displays the generated QR code for confirmation.

9. UI Policies and Widget

Catalog UI Policy: Fields Visibility

Catalog UI Policy Action: Enter Payment Mode

Service Portal Widget: Metro QR Widget

10. Project Workflow

1. User opens Book A Metro Ticket.
2. Selects Starting Station.
3. Selects Destination Station.
4. Selects Journey Type.
5. Selects Number of Passengers.
6. System calculates the applicable fare.
7. User selects Payment Mode.
8. User submits the ticket.
9. QR Code Generation is triggered.
10. Metro Ticket QR is displayed.

11. QR Ticket Generation

The project uses a Service Portal widget named Metro QR Widget. The QR generation client script creates a ticket reference URL and passes the QR image URL to the widget. The QR code is displayed to the user before final confirmation.

12. Testing

- Source and destination station selection
- Single journey fare calculation
- Return journey fare calculation
- Passenger-based fare calculation
- Payment mode selection
- Field visibility
- QR code generation
- Ticket submission

13. Sample Test Cases

Test Case 1: Ameerpet → Madhapur, Single Journey, 2 passengers. Expected fare: $30 \times 2 = 60$.

Test Case 2: Ameerpet → Madhapur, Return Journey, 2 passengers. Expected fare: $30 \times 2 \times 2 = 120$.

14. Update Set and Deployment

Update Set: Metro Ticket Generating System Project

The exported Update Set XML is maintained in the GitHub repository under the ServiceNow folder. To restore the configuration in another ServiceNow instance, import the XML through System Update Sets, preview it, and commit it after verification.

15. Repository Structure

README.md contains the project overview and usage information. Screenshots contains project demonstration images. ServiceNow contains the exported Update Set XML.

16. Future Enhancements

- Real-time metro station data integration.
- Online payment gateway integration.
- Database-based fare management.
- User ticket history.
- Ticket cancellation functionality.
- Live metro route and station information.
- Improved QR ticket validation.

17. Author

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